

Shi Hao Ng

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EDUCATION

Transfer to BEng requested; expected graduation Jun. 2028

Imperial College London

London, UK

Master of Engineering: Computing AI/ML, Year 1: Upper Second Class (2:1)

Sep. 2025 – Jun. 2029

Marlborough College Malaysia

Johor, Malaysia

IB Bilingual Diploma: 42/45 (7, 7, 7 in Higher Level subjects)

Graduated May 2025

- IGCSE: 10A* 2A. Best score in Asia for Computer Science; Best in Malaysia for Design Technology, Physics & Chemistry

EXPERIENCE

Robotics & Machine Learning Intern

July 2026 – Sep. 2026

Vikaso Robotics 4.0

Buckinghamshire, UK

- Built Vikaso's first camera-intrinsic and camera-to-robot extrinsic calibration pipeline, achieving ~3 mm initial extrinsic accuracy; deployed and tested for 100+ hours on Universal Robots UR5e arm & Intel RealSense camera
- Built Vikaso's first configurable perception stack, unifying 3 segmentation models (CAD-SAM3, CNOS, SAM-6D) and 4 6D pose estimation models (FoundationPose, MegaPose, GigaPose, SAM-6D) behind a common interface
- Built a benchmark harness evaluating integrated models on BOP 6D localization and 2D detection tasks across 10 datasets (LMO, HOT-3D, IMD, etc.). Results matched each model's published AR score, which verified the integrations reproduced the papers.

Re-Founder & Co-President

Oct. 2025 – Apr. 2026

Imperial College Drone Society

London, UK

- Re-established Imperial College Drone Society, raised £4k, reached 80+ members, ran drone-building events

Engineering Intern

Jun. 2025 – Jul. 2025

Nosco Asia (S) Pte Ltd

Taichung, Taiwan

- Assisted hot-repair ceramic-welding crews servicing still-running large-scale glass furnaces. Operated welding machinery, performed ceramic welding, cut refractory bricks, and learned how glass production lines operate

PROJECTS

Intrinsic AI for Industry Challenge | PyTorch, LeRobot ACT, Isaac Sim, Gazebo

2026

- Experimented with RL (Isaac Lab), imitation learning (ACT) and scripted FSM policies for fiber-optic cable insertion into server-rack ports on a simulated UR5e; placed 49th of 196 teams with Team Quantarune
- For the final submission, developed a hybrid ACT-FSM policy using ACT (Action Chunking Transformers) for approach and cable-to-port alignment, followed by rule-based insertion with force-scaled steps and lateral motion limits. Added automatic retraction and realignment to recover from failed attempts
- Scored 89 on held-out competition evals versus 141 locally. Later analysis found the local tests did not reflect competition conditions, motivating a paper on coverage-driven evaluation (in preparation)

GPT-2 Pre-training: Layer Normalization Study | PyTorch, DDP, HPC | GitHub | Paper

2024 – 2025

- Reproduced GPT-2 124M from scratch in PyTorch, pre-trained on FineWeb using DDP on 4× NVIDIA H100s; tuned GPU configurations, batch size, mixed precision and torch.compile, achieving training throughput of 1.27M tokens/s
- Implemented LayerNorm, RMSNorm & PowerNorm. RMSNorm achieved comparable loss at 6% higher throughput than LayerNorm; PowerNorm's gradients exploded. Ran normalization ablations and analyzed per-layer gradients
- Fine-tuned base & ablated models on BillSum and SQuAD. Evaluated text generation, summarization and QA using ROUGE/F1, Kruskal-Wallis tests and GPT-4o (LLM-as-a-judge). Published weights on Hugging Face; awarded Grade A for this IB Computer Science Extended Essay

C++ OpenGL Game Engine | C++17, OpenGL, GLFW, ImGui | GitHub | Website

2022 – 2023

- Recreated game engine in C++/OpenGL following The Chernobyl's tutorials: batched sprite renderer with texture slots over shared buffers, GLSL shaders for texturing and lighting, and a hand-written matrix library with a transformation stack
- Built an application framework using GLFW for windows & input and C++ templates to dispatch events to handlers, with a layer stack that routes events to overlays first. Integrated ImGui docking and multi-viewport support into the render loop

Augmented Electromagnetic Railgun | LS-Dyna, Fusion 360, High-Voltage Electronics | Website | Paper

2024 – 2025

- Iterated through 3 physical railgun prototypes, modeling each in Fusion 360 before fabrication. Ran 70 LS-DYNA simulations across 10 voltage conditions and decided on augmented railgun design informed by US Navy research before the final build. Cut copper rails and assembled the prototypes with acrylic panels laser-cut from Adobe Illustrator drawings
- For the power system, built a 450 V, 10 mF capacitor bank, charged via a ZVS driver, hand-wound ferrite-core transformer and bridge rectifier. Final railgun averaged 132 km/h across 10+ firing trials, estimated from high-speed footage

Custom FPV Racing Drone | Blender, Betaflight, PET-CF 3D Printing, ELRS | Website

2025

- Designed the frame in Blender and 3D-printed in PET-CF; hand-soldered electronics (Mamba F722, DJI O4 Pro, ELRS), tuned Betaflight PIDs. Replaced frame due to crash damage; flew the completed 610 g drone in Acro (240 km/h theoretical top speed)

F1 in Schools Competition | Fusion 360, Ansys Fluent, Mastercam, FDM & Resin Printing | Website | Team Site

2023 – 2024

- Team Lead & Technical Lead: secured £0.9k sponsorship; led Fusion 360 design, optimization with Ansys Fluent CFD, resin printing & CNC-machining (Mastercam); built a 20 m test track, saving the school £9.1k; won Innovation & Marketing awards

TECHNICAL SKILLS

Languages: Python, C/C++, Kotlin, Java, Haskell, TypeScript/JavaScript, RISC-V assembly

ML: PyTorch, DDP, Hugging Face Transformers, LeRobot, YOLO, SAM, Weights & Biases, NumPy, vLLM, llama.cpp

Robotics & Simulation: ROS 2, MuJoCo C API, Gazebo, Isaac Sim, FoundationPose, OpenCV, Universal Robots UR5e

Systems & Web: Linux/CUDA toolchain, Git, Docker, GCP, AWS, OpenGL, GLFW, React, Node.js, Firebase

FEA & Hardware: Fusion 360, Blender, Ansys Fluent, LS-Dyna, 3D printing, CNC, high-voltage electronics, Betaflight, soldering